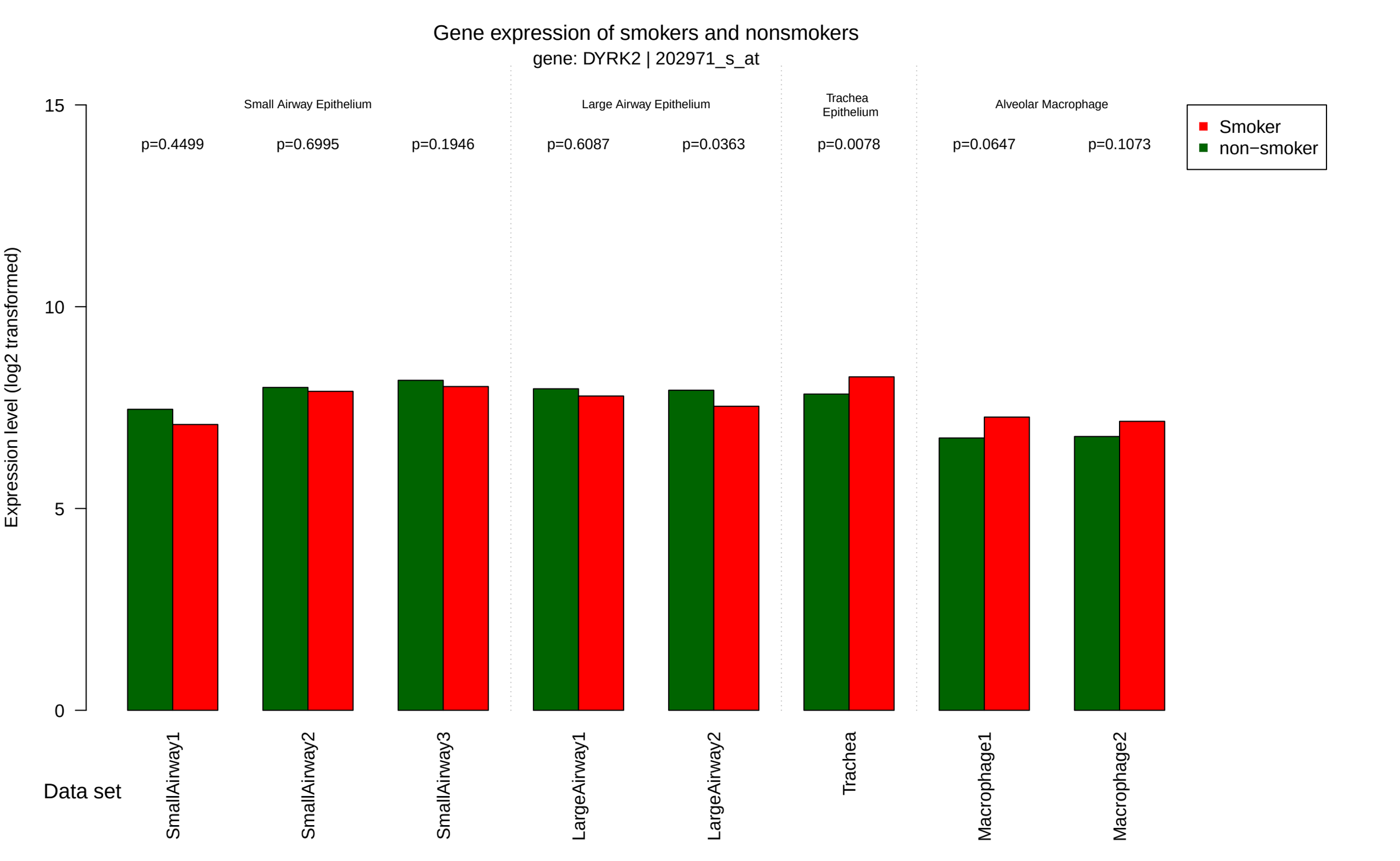
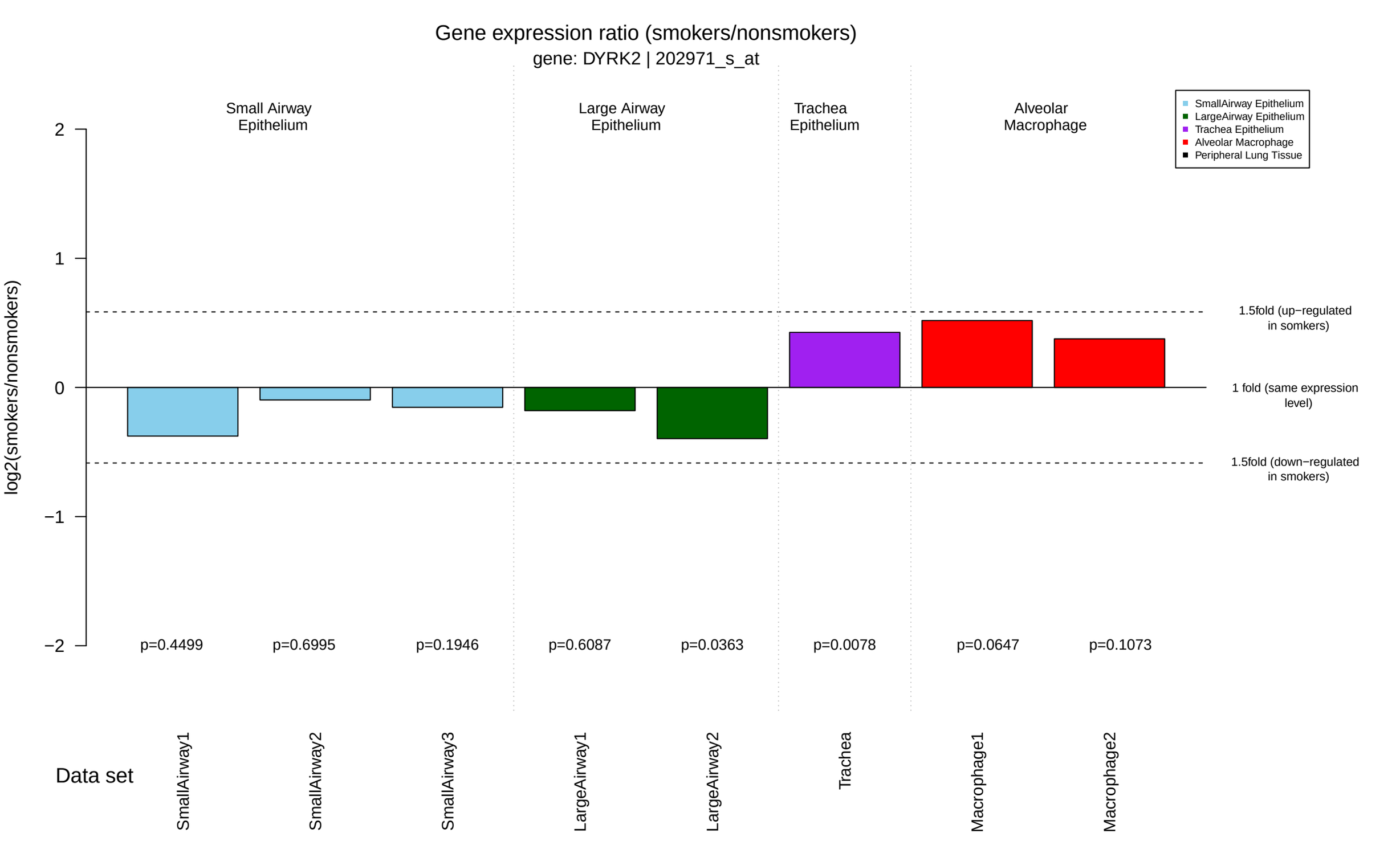
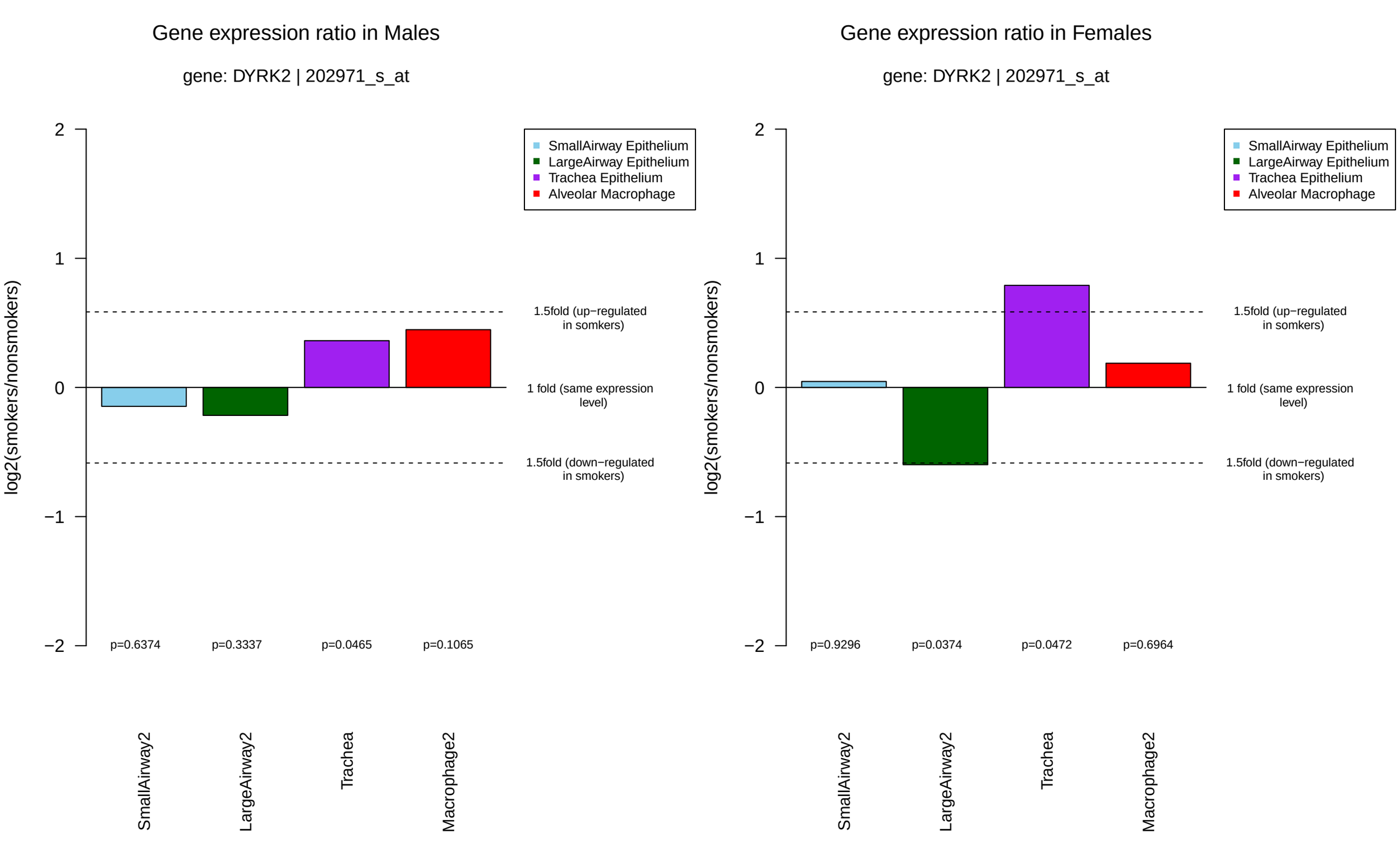
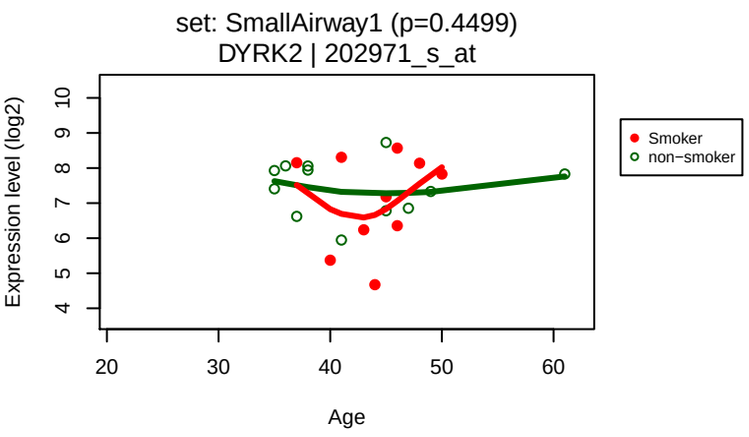
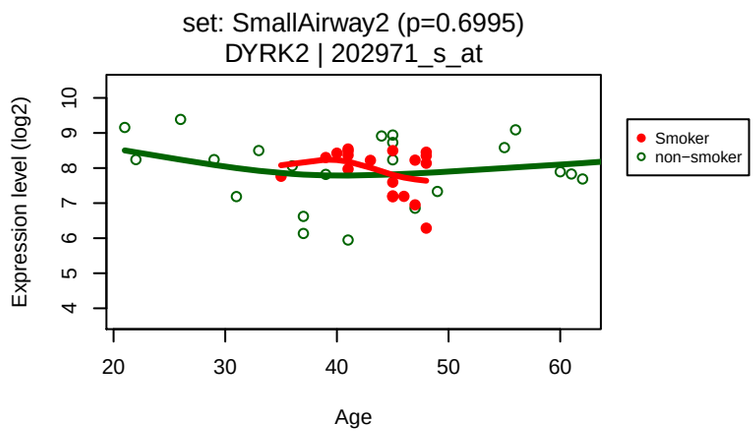
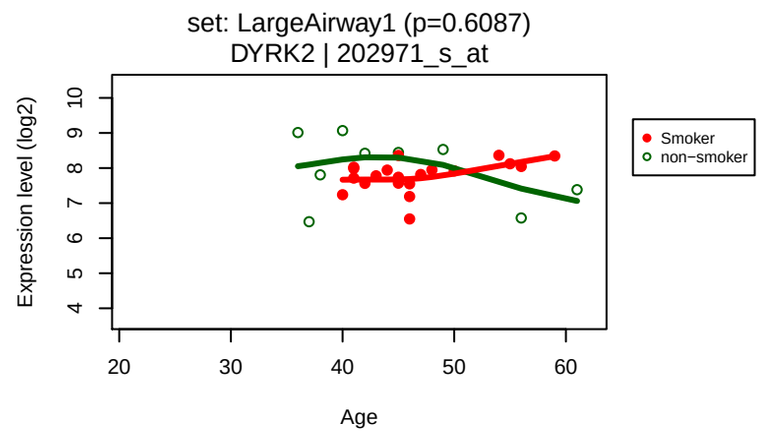

Gene expression of smokers and nonsmokers
gene: DYRK2 | 202971_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4499
p=0.6995
p=0.1946
p=0.6087
p=0.0363
p=0.0078
p=0.0647
p=0.1073
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

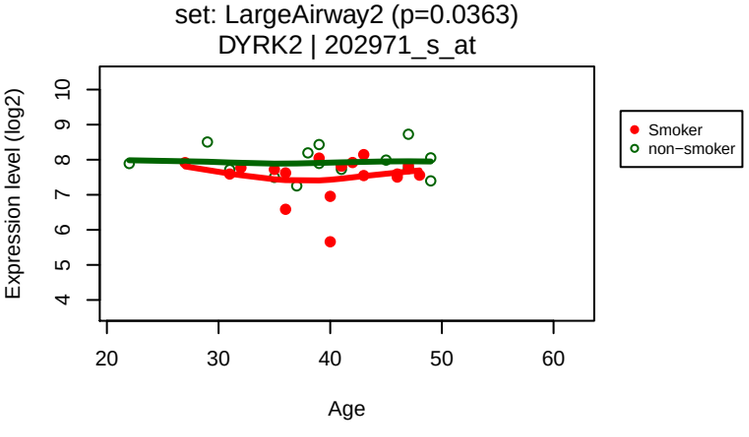

Gene expression ratio (smokers/nonsmokers)
gene: DYRK2 | 202971_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4499
p=0.6995
p=0.1946
p=0.6087
p=0.0363
p=0.0078
p=0.0647
p=0.1073
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

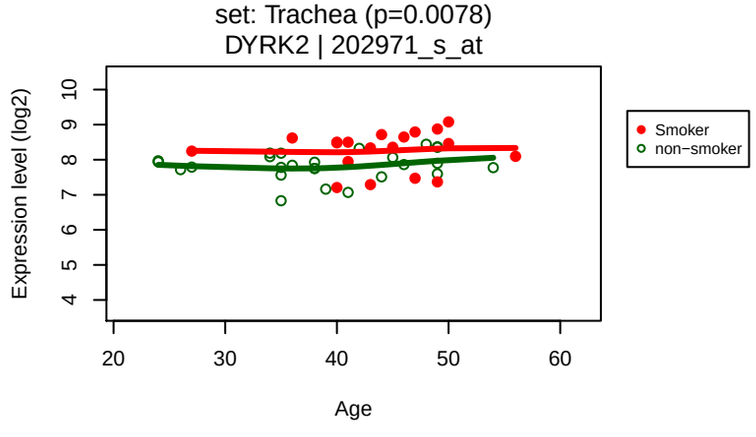

Gene expression ratio in Males
gene: DYRK2 | 202971_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6374
p=0.3337
p=0.0465
p=0.1065
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DYRK2 | 202971_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9296
p=0.0374
p=0.0472
p=0.6964
SmallAirway2
LargeAirway2
Trachea
Macrophage2

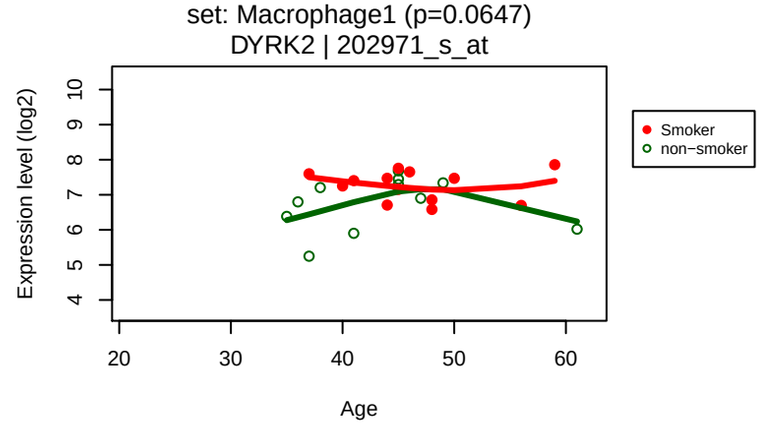

set: SmallAirway1 (p=0.4499)
DYRK2 | 202971_s_at
Expression level (log2)
Age
Smoker
non−smoker

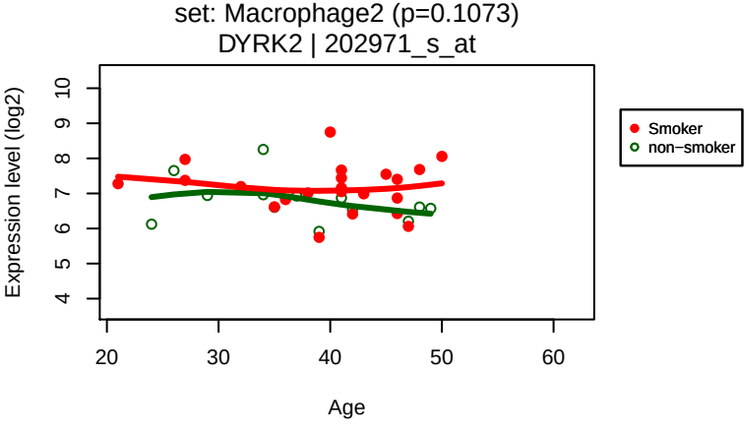

set: SmallAirway2 (p=0.6995)
DYRK2 | 202971_s_at
Expression level (log2)
Age
Smoker
non−smoker

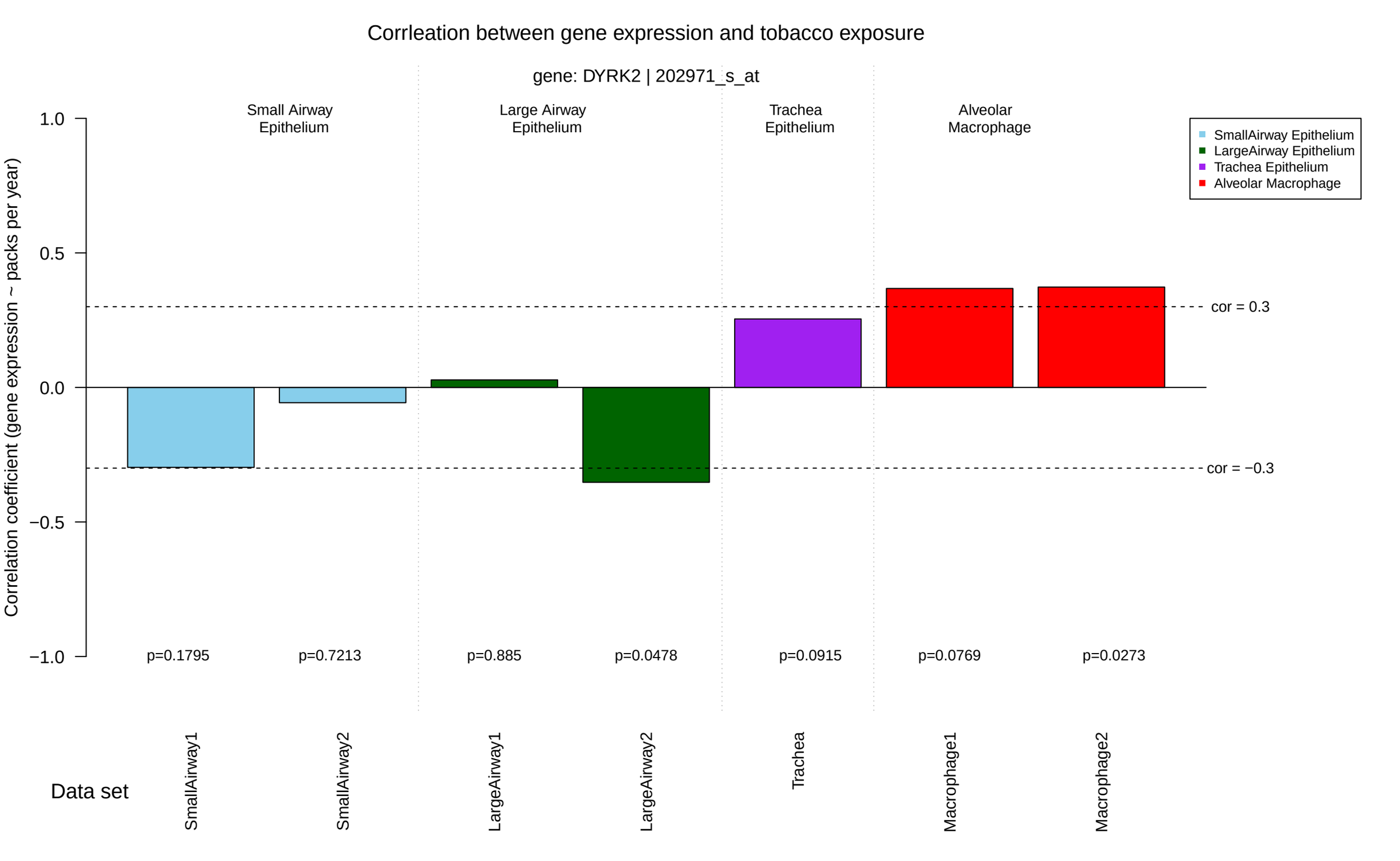

set: LargeAirway1 (p=0.6087)
DYRK2 | 202971_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0363)
DYRK2 | 202971_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0078)
DYRK2 | 202971_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0647)
DYRK2 | 202971_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1073)
DYRK2 | 202971_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DYRK2 | 202971_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1795
p=0.7213
p=0.885
p=0.0478
p=0.0915
p=0.0769
p=0.0273
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2